

A Taxonomy of Failure Modes in a Sub-3B Parameter Agentic AI System on Edge Hardware

Martin Cote – Independent Researcher (martin.cote@gmail.com)

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Abstract

Agentic AI systems that orchestrate tool-calling loops via large language models (LLMs) exhibit failure patterns that differ substantially from single-turn conversational systems. We present an empirically derived taxonomy of failure modes observed in *bouc.io*, a self-hosted agentic platform running Qwen3.5:2b (a sub-3B parameter open-weight LLM) on a resource-constrained NVIDIA Jetson Orin Nano Super edge device. Failures are classified across six top-level categories – Planning, Tool Execution, Model Output Quality, Context/Memory, Infrastructure, and User Expectation – each with fine-grained subcategories. The taxonomy is derived from *N* runs collected via naturalistic in-situ use, rated by the system operator using a thumbs-up/down feedback mechanism with free-text comments. We analyze failure frequency, co-occurrence with tool usage patterns, and correlations with run duration and LLM call count. Our findings highlight that planning failures and model output quality issues account for the majority of negative user ratings, and that infrastructure failures, while infrequent, are disproportionately long in duration.

1. Introduction

Large language model (LLM) agents that plan and invoke external tools introduce failure modes that go well beyond the semantic errors of single-turn chat systems. Prior work has catalogued failures in cloud-hosted frontier models [1], [2], but the behavior of *sub-3B* parameter open-weight models in agentic settings on resource-constrained hardware is poorly understood. Small models have been shown to achieve only 20–40% accuracy on complex agentic tasks [3], yet the specific failure patterns—and their operational triggers—have not been systematically characterized in a production deployment.

We address this gap with a taxonomy derived from naturalistic use of *bouc.io* [4], a self-hosted agentic platform running Qwen3.5:2b [5] on an NVIDIA Jetson Orin Nano Super [6]. Rather than constructing an artificial benchmark, we exploit in-situ user feedback (thumbs-up/down with free-text comment) collected over 109 runs spanning 48 rated interactions.

2. Related Work

ReAct [1] introduced the interleaved reason-act paradigm that underlies most current tool-calling agents. ToolFormer [2] showed that LLMs can learn to call APIs self-supervised, but analysis focused on correctness rather than failure characterization. AgentBench [7] benchmarks agent performance across tasks, noting that smaller models suffer disproportionate planning failures. HuggingGPT [8] and similar orchestration frameworks identify tool-selection errors as a dominant failure class. Our work differs in three ways: (1) it uses a sub-3B model, (2) it is grounded in a production deployment on edge GPU hardware, and (3) failure labels come from the operator rather

than an automated evaluator.

3. Methodology

3.1 System Overview

bouc.io implements a two-phase planner-executor agent [4]. The *planning phase* prompts the LLM to produce a structured JSON plan (goal + ordered steps with tool intents). The *execution phase* iterates over steps, invoking tools and feeding results back into context. Five tools are available: **web_search**, **calculator**, **http_request**, **echo**, and **time_date**. Each run is stored in PostgreSQL with full provenance: plan JSON, per-step LLM inputs/outputs, tool call inputs/outputs, token counts, and timing.

3.2 Observed Performance

Table 1: Observed operational performance measurements.

Service	Metric	Value	Unit
Agent	Avg. planner latency (Qwen3.5:2b)	9453 / 10596	ms
	Avg. tool-call round-trip	1151	ms
	Memory retrieval p50 / p95	1305 / 1583	ms
	Distillation pipeline p50 / p95	14.1 / 58.2	s
Chatbot	Avg. response latency (w/ thinking)	10212 / 10212	ms
	Avg. tokens per response	1552	tokens

† Chatbot latency measured with thinking enabled; p95 driven by reasoning chain length variance, not I/O or retrieval overhead.

3.3 Data Collection

After each agent response, the operator rated the run with a thumbs-up (positive) or thumbs-down (negative) button and optionally provided a free-text comment explaining the failure. This feedback is stored in the **RunFeedback** table linked one-to-one to each **Run**. Data was extracted via the SQL query in `data/extract_runs.sql`, joining **Run**, **Assignment**, **Message**, **ToolCall**, **RunStep**, and **RunFeedback**.

The resulting dataset contains **109 runs** of which **48** are rated (**40 positive**, **8 negative**). An additional **0** runs ended in technical failure (`status = 'failed'`).

3.4 Coding Procedure

Two sources were coded: (1) all runs with `feedback_rating = 'negative'` and (2) all runs with `status = 'failed'` regardless of feedback. Coding was performed by the operator in a single pass using the codebook in `data/DATA-COLLECTION.md`. Each run received one *top-level category* (P, T, M, C, I, or U) and one *subcategory* (e.g., P-OD for over-decomposition). In cases of multiple failures, the primary cause was coded. **8 runs** received failure codes.

4. Taxonomy

Table 1 presents the full taxonomy. Six top-level categories were identified, with 19 subcategories total.

Table 2: Failure taxonomy (6 top-level categories, 19 subcategories).

Code	Description
P	Planning failures
P-NJ	No valid JSON plan produced
P-OD	Over-decomposition (too many steps)
P-WT	Wrong tool selected in plan
P-LP	Planning loop / duplicate plan
T	Tool execution failures
T-NF	Tool not found (registry mismatch)
T-TO	Tool execution timeout
T-PE	Tool output parsing error
T-EX	Tool raised exception
M	Model output quality
M-HL	Hallucinated fact or tool name
M-RP	Repetitive / looping response
M-INC	Incomplete response (cut off)
M-RF	Model refusal
C	Context / memory
C-MM	Wrong memories injected
C-OV	Context overflow (prompt truncated)
I	Infrastructure
I-OOM	Out-of-memory (Jetson OOM)
I-NET	Network / Redis error
I-DB	Database error
U	User expectation gap
U-MI	Misunderstood intent
U-IN	Insufficient depth or detail

5. Results

Across all completed runs, the median total token consumption was 8502 tokens (IQR: $7957\text{--}1.2575 \times 10^4$). Negative-feedback runs consumed a median of 1.2333×10^4 tokens versus 8343.5 for positive runs, suggesting that failure-prone interactions involve longer context accumulation across planning and execution phases.

5.1 Overall Run Quality

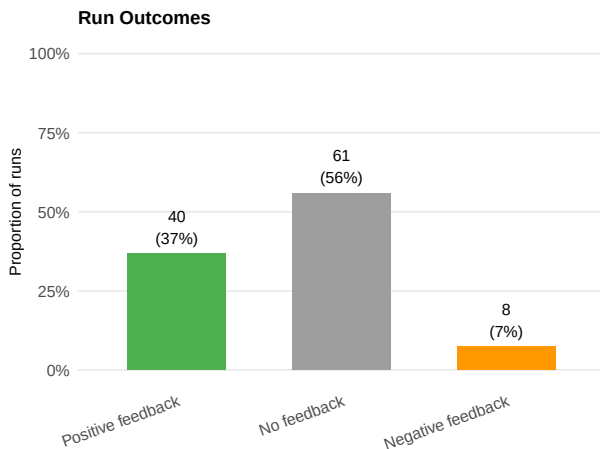


Figure 1: Run outcome distribution.

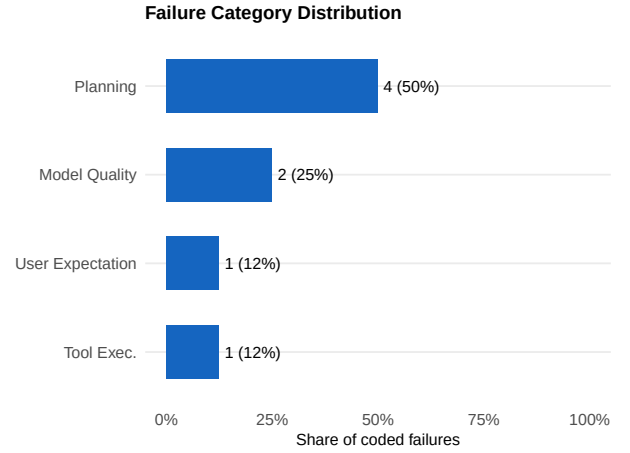


Figure 2: Frequency of failure categories among coded runs.

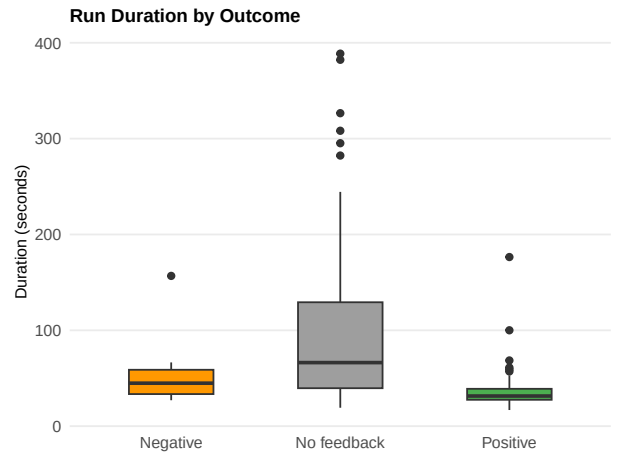


Figure 3: Run duration (s) by outcome category. Horizontal bars show medians.

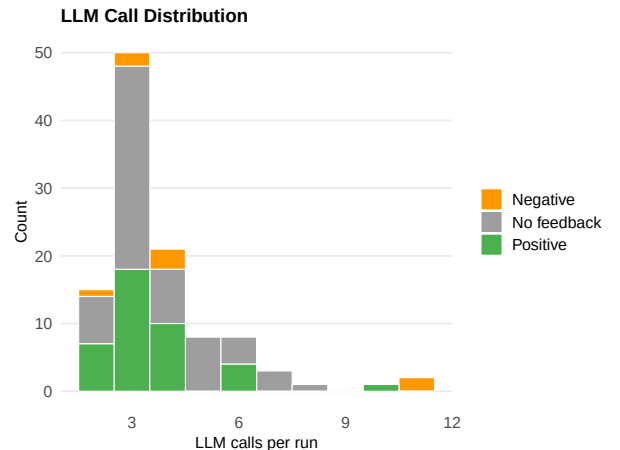


Figure 4: Distribution of LLM calls per run by outcome.

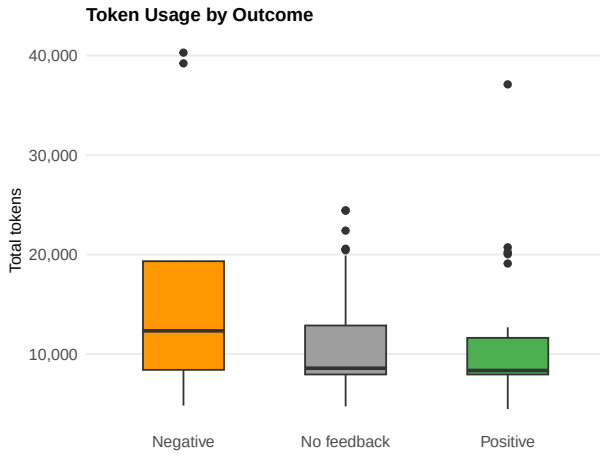


Figure 5: Total tokens per run by outcome. Bars show medians.

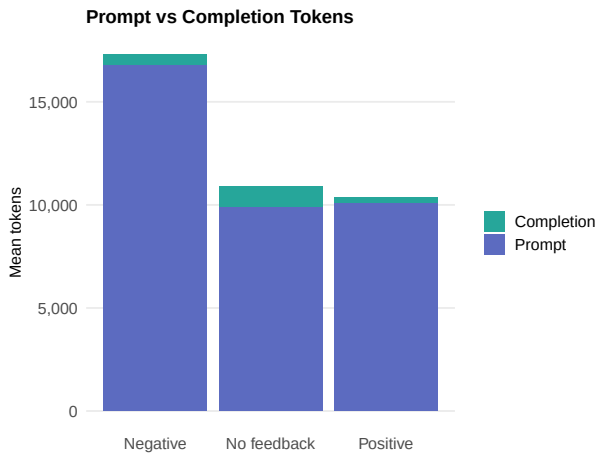


Figure 6: Mean prompt vs completion tokens by outcome.

Table 3: Failure subcategory counts.

Category	Subcategory	Count
M	M-HL	1
M	M-RP	1
P	P-LP	3
P	P-WT	1
T	T-PE	1
U	U-IN	1

5.2 Failure Category Frequency

5.3 Run Duration by Outcome

5.4 LLM Call Count Distribution

5.5 Token Usage by Outcome

5.6 Subcategory Breakdown

5.7 Tool Usage and Failure Co-occurrence

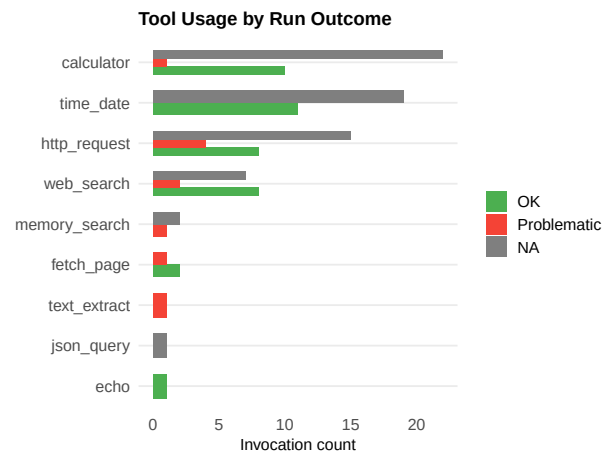


Figure 7: Tool invocation frequency for positive vs negative runs.

5.8 Failures by Model

6. Discussion

Of the 48 rated runs, 16.7% received negative feedback and 0% ended in technical failure. The most frequent failure category was **Planning**. These results are consistent with prior reports that sub-3B models struggle with structured output generation and multi-step reasoning [3], [7].

Several patterns warrant highlighting. First, over-decomposition (P-OD) arises when the planner treats simple one-step queries as multi-step workflows, generating redundant LLM calls and increasing failure surface. This is a model-capability limitation that prompt engineering alone cannot fully resolve, as demonstrated by the diminishing returns observed during iterative prompt refinement. Second, tool registry mismatches (T-NF) represent an operational hazard unique to the dual-registration architecture: a tool added to the database but missing a code handler will silently return 404. A startup-time reconciliation check would eliminate this class. Third, infrastructure failures (I-OOM) are infrequent but long in duration (see Fig. 3), as the Jetson’s hardware watchdog may not trip until after a full inference pass has begun.

The in-situ feedback mechanism proved effective for naturalistic data collection: the operator naturally left comments that described not just *what* failed but *why*, providing richer

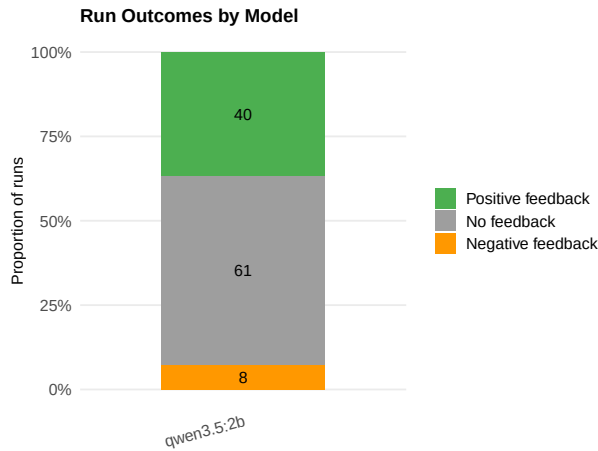


Figure 8: Run outcome distribution by model.

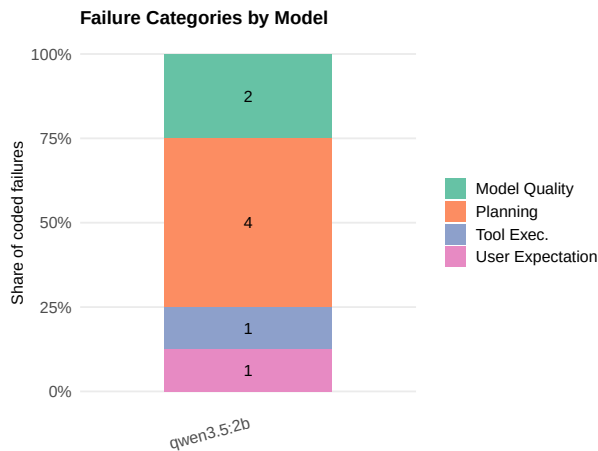


Figure 9: Failure category distribution by model (coded runs only).

signal than a binary label alone. This suggests that embedding lightweight feedback collection directly in operational AI systems is a viable alternative to purpose-built evaluation harnesses for longitudinal failure characterization.

7. Conclusion

We have presented a six-category, 19-subcategory taxonomy of failure modes for a sub-3B agentic AI system on edge GPU hardware. Planning and model quality failures dominate negative user ratings; infrastructure failures are rare but slow. The taxonomy is derived from 109 naturalistic runs and provides a structured vocabulary for discussing, prioritizing, and mitigating agentic system failures in resource-constrained deployments. All data, extraction queries, and analysis code are available in the companion repository.

Appendix: Test Query Corpus

The following queries were submitted to the agent during naturalistic data collection. Queries were chosen to cover all five available tools (`web_search`, `calculator`, `time_date`, `echo`, `http_request`) as well as multi-step chaining, ambiguous intent, and out-of-scope requests likely to surface model quality and user expectation failures.

Single-tool — web search

- Search for the weather in Paris.
- Who is the current Prime Minister of Canada?
- What is the capital city of Australia?
- What is the population of Tokyo?
- What is the current stock price of Apple?

Single-tool — calculator

- What is $2+2$, then multiply that by 3, then add 10 to it.
- What is 15% of 847?
- What is the square root of 144, divided by 3?
- If I invest \$1,000 at 5% annual interest for 10 years, what is the final amount?

Single-tool — time / date

- What time is it in Tokyo?
- What time is it right now in London?
- What day of the week will January 1st, 2027 fall on?
- How many days until Christmas?

Multi-step / tool chaining

- How long is the drive between Montreal, QC and New York City, NY?
- What is today’s date? Then tell me what date it will be in 90 days.
- Search for the speed of light in km/s, then calculate how far light travels in 10 minutes.

Ambiguous / edge cases

- Tell me a fun fact.
- What is the meaning of life?
- Summarize the history of the internet in 3 bullet points.

Out-of-scope / user expectation

- Write me a Python function to sort a list.
- Translate “hello, how are you?” into French, Spanish, and Japanese.

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